STUDY MATERIAL FOR TBL 9.3.

SUMMARY MATERIAL FROM CHAPTER 16, "Biología Celular del Cáncer", Biología Celular Biomédica

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Learning objectives:

- To know the impact that cancer has in today's society, its epidemiology and its principal risk factors.
- To know main characteristics of a tumor cell, which differentiates it from the rest of the cells in an organism, and the process of tumor development.
- To study the genetic basis of cancer.
- To understand the role of some viruses in cancer development.
- To study the classic treatment against cancer and the new drugs directed towards therapeutic targets, developed mainly thanks to the study of the cancer cells.

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Introduction

Cancer is a genetic disease that alters the delicate equilibrium that exists between proliferation, apoptosis and cellular differentiation. Tumor cells proliferate without control and undergo numerous changes that affect the regulation of the cell cycle, apoptosis, adhesion, movement, etc. These changes take place in a progressive manner and are accompanied by an increasing number of genetic modifications, which provide these cells with evolutionary advantages, and allow their dissemination throughout the organism and resistance against unfavorable conditions and drugs. Not all types of cancer originate and evolve in the same manner. In reality, what we call "cancer" encircles a vast variety of diseases with some similar characteristics, and with specific differences. Some inherited mutations, genetic and epigenetic changes produced by carcinogens, certain viral infections or chronic inflammatory processes, which alter the normal physiology of a tissue, are causes that can make a normal cell to transform into a tumor cell. The cell then stops being susceptible to the regulatory signals of the equilibrium between cell division and cell death, and starts proliferating uncontrollably, forming a tumor.

These series of events, which will be explained further down this chapter, are extremely complex, and although the knowledge about the cancer cell biology has been growing exponentially in the last few years, we are still far behind from fully understanding and dominating this disease. What is clear is that the study of the cell biology of cancer allows, on one hand, the development of more effective and specific therapies directed toward the altered pathways in some specific cancer types; and on the other hand, to better understand how a tumor cell functions.

Risk Factors

Cancer is a disease that does not have a unique cause. Distinct factors can contribute to the development of the tumor. These factors vary depending on the type of cancer. The main risk factors are age, gender, genetics, diet and exposure to carcinogens. In general, the incidence of cancer increases with age, which is understood due to the accumulation of cellular alterations throughout a person's life. The increase of life expectancy has resulted in higher cancer incidence.

Lastly, diet and lifestyle are greatly responsible for the development of cancer. It has been shown that diets poor in fiber and alcohol consumption are risk factors for colorectal cancer (CRC). Also, alcohol and tobacco, many times consumed together, act in a synergistic manner, favoring the development of lung, head and neck tumors. Tobacco has a clear relation with various types of cancer, either by direct contact, such as tumors in the pharynx, larynx, stomach or lungs, or by indirect contact, such as pancreas and bladder tumors. The exposure to some other substances can also promote the development of cancer.

Characteristics of Tumor Cells

Cancer cells are characterized by:

• **High proliferation potential**: tumor cells are able to maintain an autonomous and uncontrolled cellular replication due to the alteration of the proliferation signals. This is due to altered expression of growth factors, constitutive activation of intracellular signaling pathways and paracrine/autocrine stimulation.

• **Unlimited Replication Capacity**: in non-tumor cells, the process of mitosis causes a progressive shortening of the telomeres. After certain number of replications, the cells lose their capacity to undergo mitosis. However, tumor cells express the enzyme telomerase, which adds repetitive telomeric DNA sequences and maintains a constant longitude of the telomeres. The cell, thus, acquires an unlimited replication capacity, without entering into senescence.

• **Blockade of apoptosis**: Tumor cells block apoptosis by increasing expression of antiapopototic molecules, such as bcl-2, and blocking the expression of proapoptotic genes, like Bax.

• **Capacity of Invading and Metastasizing**: in advanced tumors, the adhesion between neighboring cells is decreased due to the acquisition of a migratory mesenchymal phenotype and a lower expression of adhesion molecules, such as E cadherin. This allows these cells to move and invade nearby tissues and/or metastasizing to distant organs.

• **Induction of Angiogenesis:** the tumor growth must acquire an angiogenic phenotype that induces the formation of new blood vessels in order to provide oxygen and nutrients and elimination of carbon dioxide and waste products. The lack of neovascularization will lead to necrosis of the tumor.

• **Evasion from the immune system:** tumors are able to escape from the effect of the immune system, in particular from B- and T-lymphocytes, macrophages and NK cells.

Types of Cancer

Tumors (or **neoplasms**) can be classified as follows: According to their behavior, as benign or malignant. **Benign** tumors present a slow growth rate and a high degree of differentiation, are located in precise areas and are not extended into other areas; in other words, they do not produce metastasis. **Malignant** tumors are undifferentiated and show a high growth rate. They can cross the basal membrane, invade local tissues and cause metastasis by spreading through a lymphatic, blood or serous pathways. A different classification criteria relates to the tissue they derive from. Tumors can derive from epithelial, mesenchymal, lymphoid, nervous or germinal cells. In order to name a tumor, the root word of the tissue of origin, followed by the suffix –oma is used. Benign tumors coming form epithelial cells are called papillomas or polyps, depending on whether they originate from a stratified or cylindrical epithelium, respectively. Those derived from glandular tissue are known as adenomas. Epithelial tumors with malignant characteristics are known as **carcinomas** and the malignant variant of the adenoma is the **adenocarcinoma**. Non-epithelial malignant tumors are known as **sarcomas**. Lastly, tumors can be classified by the organ or system they originate from: lung, liver, prostate, breast, colon/rectum, bladder, hematological tumors (leukemia, lymphoma), etc.

Induction of Cancer by Carcinogens and Microorganisms

Carcinogenesis refers to the process of cancer development. It consists of various stages where mutations are produced and clonal selection preserves those cells with higher proliferation capacity, survival, invasion and metastasis. The genesis of cancer, although not completely clarified for many tumor types, usually begins with one or multiple genetic alterations that confer hyperproliferative ability. These alterations can be inherited or acquired. In the latter case, they can be produced by carcinogenic agents, by genetic anomalies caused by microorganisms or by the alterations of cell signaling.

Physical Carcinogens

Carcinogens of physical origin are mainly radiations we are exposed to, which can be further classified into: a) **non-ionizing radiation**, such as solar radiation, and b) **ionizing radiation**, such as X-rays. Both types of radiation cause DNA damage that must be repaired by cells before mitosis takes place. Otherwise, the alterations could be transmitted to daughter cells and favor the process of malignant transformation.

Ultraviolet rays from sun light belong to the group of non-ionizing radiation. In normal conditions, cells muffle these damages through DNA repair mechanisms. However, if there is a defect in these mechanisms, the damage cannot be repaired. UVA rays cause indirect damage to the genetic material, by generating **free radicals** that can oxidize (and damage) the DNA. UVB rays cause direct damage to the DNA through the production of covalent bonds between adjacent pyrimidines of the same DNA strand (**pyrimidine dimers**). These dimers are eliminated by the NER (nucleotide excision repair) mechanism, which uses more than repair 30 proteins. The dimers that escape this repairing process can generate mutations that, when affecting genes involved in the cell cycle, can lead to the malignant transformation of cells. Melanoma is associated with the appearance of TT dimers in tumor cells (as a consequence of UV light) and these dimers constitute a genetic footprint of this tumor.

The exposure to ionizing radiation is clearly another inducing factor of carcinogenesis. These radiations produce DNA oxidation and breaks, thus incrementing cancer risk. A malignancy that develops frequently in the first years after exposure to ionizing radiation is leukemia. This type of radiation can also cause solid tumors, whose risk of appearance increases with time. The effects of the nuclear bombs in Hiroshima and Nagasaki are widely known, where a high cancer incidence in the population that survived was observed. The

same was found in patients affected by the nuclear accident in Chernobyl.

Chemical Carcinogens

Our organism is exposed to a number of increasing chemical compounds that have a diverse degree of carcinogenic potential. In many cases this potential is not found until many years after exposure to these carcinogens. The first observations about the carcinogenic potential of some chemical substances were the high rates of testicular cancer found in chimney sweeps. Later, it was found that the exposure of workers to asbestos, cadmium and other chemical elements also favored the development of tumors.

Tobacco contains a high number of components with carcinogenic potential, such as tars, polycyclic hydrocarbons or nitrosamines. Some of these carcinogens damage the DNA and alter critical genes that promote tumor development. Also, they trigger an intense inflammatory response. Because of the continuous inhalation of these substances, **inflammation becomes chronic**, with production of large amount of reactive species of oxygen that damage DNA, as well as cytokines and growth factors that stimulate the proliferation of tumor cells.

Microorganisms that induce carcinogenesis

The importance of certain infections on the appearance of some types of cancer is now well recognized. Investigations in this line have been fundamental in order to establish prevention programs for these tumors. Among the pathogenic microorganisms, viruses have an essential role. The important carcinogenic role of some viruses is explained in the section on "oncogenic viruses" of this summary.

Some bacteria also possess carcinogenic potential. Probably the most remarkable case is ***Helicobacter pylori***, a bacterium discussed in chapter 1. It is estimated that about 30% of the healthy population has gastric mucosa colonized by H. pylori. However, not all of these individuals will present clinical manifestations. Its presence in the gastric mucosa causes a chronic superficial gastritis, which leads to an abnormal increase in secretion of hydrochloric acid. In this context of chronic inflammation, inflammatory cells produce free radicals that can cause DNA damage.

Genetic Basis of Cancer

As we have mentioned, the appearance of cancer has a genetic basis over which environmental factors can influence and favor the development of this disease. We will now discuss the different genetic factors that can cause cancer.

Alterations in DNA repairing

Damage caused in the DNA by external carcinogenic (radiation or chemical agents) or endogenous (free radicals derived from cellular metabolism) agents, and errors produced during replication must be repaired in order to maintain the integrity of the genome. Any change in the structure of the double helix is known as DNA damage. Within this category are included phosphodiester bond ruptures, elimination of bases, incorporation of reactive covalent groups, conversions of one base to another and introduction of covalent bonds between bases within the same DNA strand. This activates genes that produce a standstill in the cell cycle, with the goal to repair those injuries. If this is not possible, the cell induces its own death through apoptosis. The alteration of this repair system can produce cancer.

In some of the repair mechanisms, the DNA lesion is recognized by a series of XPC proteins that will facilitate the removal of a small piece of the affected DNA strand. This space is filled in by DNA polymerase δ and ε, and lastly, the DNA ligases seal both ends of the DNA together. The alteration of this repair mechanism gives rise to a disorder of recessive autosomal origin known as **Xeroderma pigmentosum** (XP) (**fig. 1**). Patients with XP are born with skin that appears normal. However, from the first months of life they begin to present reversible alterations on their skin, such as erythema and peeling, which worsen with sun exposure. In fact, a brief exposure to the sun can produce severe skin burns. In posterior stages, cutaneous atrophy and telangiectasia (capillary dilations of the skin) appear. Later on, these patients develop cutaneous tumors, such as squamous carcinomas, basocellular carcinomas and melanomas. These type of tumors, typically found in adults, appear in 4-5 year-old children with XP. They can also suffer from ophthalmological and neurological problems. This genetic disease does not have a cure. Therefore, it is important to take preventive measures, such as avoiding sun exposure and monitor the patient closely.



Fig. 1. Pacient with Xeroderma pigmentosum.

Oncogenes

Concept and generation of oncogenes

Oncogenes are defined as altered genes that promote tumor development, whose non-altered version is known as **proto-oncogenes**. They are genes that are expressed in a regulated manner in normal cells, in charge of stimulating growth and cellular division. When mutations take place (or another type of alterations, as we will see further down this chapter), whose result is an abnormal cell proliferation, one refers to **oncogenes**. Proto-oncogenes code for proteins that constitute growth factors, growth

factor receptors, components of intracellular signaling pathways, transcription factors, genes that regulate apoptosis or proteins that intervene in cellular differentiation. Oncogenes are **dominant genes**; this means that mutation in one of the alleles of the proto-oncogene is sufficient for the abnormal proliferative effect.

The formation of oncogenes can take place through the following processes (**fig. 2**):

• Through mutations that produce hyperactive proteins with a higher functionality than its native version.

• Through genetic amplification. For example, about 25-30% of breast tumors possess multiples copies of the gene Her-2, a membrane receptor whose activity promotes mitosis. In this case, the gene is not mutated, but there is an excessive genic dose with positive signals for cellular division.

• Abnormal rearrangements of genetic material. This can bring about a gene being translocated to a place of the genome where it is constantly being transcribed, or to the generation of fusion abnormal proteins that activate the cell cycle. The integration of viral DNAs into the host genome can also cause mutagenesis.

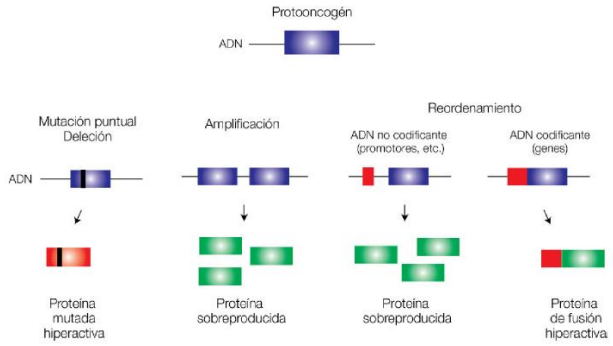


Fig. 2. Mechanisms to generate oncogenes.

Types of oncogenes

Oncogenes can belong to, at least, one of these categories:

Growth Factors – the expression of growth factors in tumor development can be seen abnormally augmented, causing hyperstimulation of cell the proliferation.

Growth Factor Receptors – these proto-oncogenes encode membrane proteins with an extracellular portion that binds to the ligand, and an intracellular domain that activates the signaling cascade. The alteration of this type of proto-oncogenes takes place due to amplification of the gene, by mutations that cause constitutive activations or by hyperactivation of the regulatory region. The hyperactivation of the epidermal growth factor receptor (EGFR) and Her-2 are examples of this type of oncogenes. EGFR signaling favors processes implicated in the growth and progression of tumors, such as proliferation, angiogenesis, inhibition of apoptosis, invasion and metastasis. The excess of this signaling is critical for the growth of many tumors. The overexpression of EGFR in diverse tumors (80-100% in head and neck tumors, 40-80% in lung tumors, 30-50% in pancreatic tumors) confers worse prognosis and worse response to conventional chemotherapy.

Genes of Intracellular Signaling Pathways – Proteins of the intracellular signaling cascade can also be altered: in non-tumor cells, the **RAS** GTPase activates the intracellular signaling pathways known as **MAPK**. RAS, the first human oncogene discovered, participates in the signaling mediated by growth factors. KRAS mutations that cause a gain of function (found in approximately 20-30% of human tumors) produce a constitutive activation of the cell cycle, favoring the development of neoplasms.

Some tyrosine kinase-type proteins implicated in the intracellular signaling pathways can also be altered. This occurs in **chronic myeloigenous leukemia**, where a **translocation of the proto-oncogene Abl from chromosome 9 to chromosome 22** is produced. The abnormal chromosome 22, denominated **Philadelphia chromosome** produces a fusion protein of the genes Bcr-Abl, with a constant tyrosine kinase activity that promotes uncontrolled proliferation.

Genes Related to Apoptosis – Some proto-oncogenes are not related to dysregulation of cell division, but with an increase of survival signaling, which translates in a greater life expectancy for altered cells. This type of genetic modification is produced, for example, in **follicular lymphoma**. In this case, the **translocation of chromosomes 14 and 18** causes the overexpression of the **bcl-2** anti-apoptotic protein.

Tumor Suppressor Genes

Tumor suppressor genes (TSG) code for proteins that inhibit or stop the cell cycle. The loss of functionality of these genes through deletions, point mutations, or epigenetic silencing has, as a consequence, a lack of negative signals necessary to control the cell cycle, which can result in uncontrolled proliferation (**fig. 3**).

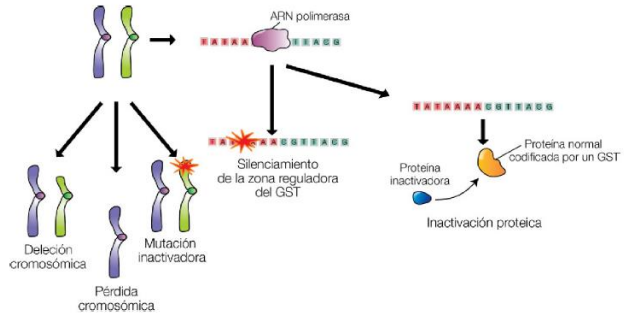


Fig. 3. Mecanisms that alter proteins encoded by TSG.

TSG are **recessive**, because they normally require the mutation or silencing of both alleles to cause cancer. If the alteration is only found in one of the two alleles, normally, the cell continues being

able to control the cell cycle due to the non-mutated allele. The number of TSG found in cancer is lower than that of oncogenes, because of their recessive character. Examples of TSG are the retinoblastoma (Rb), p53 and BRCA1. p53 gene, known as the gatekeeper of the genome, is the TSG that most frequently appears mutated in cancer. We will focus on p53 and Rb.

• **Rb** Suppressor Gene: As described in chapter 14, in its phosphorylated state, the Rb protein is released from to the E2F transcription factor, and replication initiates. The inactivation of both Rb alleles causes the protein to be unable to perform its normal function of stopping cellular proliferation. Rb was the first TSG discovered, studying families with retinoblastoma, an infrequent eye tumor. 40% of retinoblastomas have a hereditary character and appear during childhood. In these cases, children inherit a mutated allele from one of their parents and present a predisposition to developing a tumor. The inactivation of the unaffected allele through mutation (or any other type of silencing) triggers the neoplastic transformation. This phenomenon occurs frequently during embryonic development or childhood, since the majority of children that inherit a mutated copy of Rb develop retinoblastoma. The remaining 60% of retinoblastomas occur due to independent sporadic inactivations in both alleles. It is not well understood why the loss of Rb produces such premature tumors in the retina and not in other organs. Nevertheless, children diagnosed of retinoblastoma have a higher risk of suffering from sarcomas. The loss of Rb is also characteristic in some adult tumors.

• **p53**: the p53 gene produces a 53kDa protein that controls the cell cycle. In stressful conditions, levels of p53 increase and the cell cycle is halted in order to repair the DNA damage. p53 induces the expression of protein p21, a powerful inhibitor of cyclin D/CDK4, among others cell cycle complexes. This allows the blockade of the progression of the cycle, with the purpose to give time to repair the DNA. If this is not possible, p53 induces programmed cell death through promoting the expression of proapoptotic genes, such as Bax, and inhibiting that of the antiapoptotic gene bcl-2. When p53 is altered (frequently due to mutations), DNA repair is not produced and daughter cells may accumulate genetic anomalies. Mutations in p53 are mainly found in the DNA binding domain. p53 is considered the most important TSG, since it is found mutated in 50-55% of all types of human cancers. There are individuals who inherit an altered copy of p53 and present a high risk of developing cancer when the second copy of the gene becomes inactive. These individuals suffer from the **Li-Fraumeni syndrome**, where there is a tendency for developing various types of tumors, mainly sarcomas, breast cancer, glioblastoma and adrenal gland cancer. The risk of suffering from any other type of cancer is 50% at the age of 30, and it rises to 90% by the age of 70.

Epigenetic Alterations

The neoplastic transformation can be produced due to an alteration at epigenetic level. Epigenetics refers to changes in the patterns of gene expression that do not depend on modifications in the normal sequence of nucleotides, but on the covalent and reversible changes of DNA and histones. **DNA methylation** consists in the addition of a methyl group to the 5' carbon of the cytosine that is part of the CpG dinucleotide (cytosine-phosphate-guanine) through the enzymes DNA methyltransferases (DNMT) (see chapter 7). The absence of methylation is a requirement for active transcription; thus, the main consequence of DNA methylation is **silencing of the expression of a particular gene**.

Tumor cells can present abnormal methylation patterns in promoter regions of genes, leading to loss of function. Surprisingly, this methylation is at least as frequent as mutations of deletions in TSG. DNA methylation predisposes also for the appearance of mutations. This occurs because the 5mC tend to be mutagenic, causing C-T transitions. If this occurs in a TSG and it is not repaired, mutations and can cause the beginning of the malignant transformation of cells. Thus, up to 50% of point mutations in p53 are produced by 5mC.

Modification of histones constitutes another important epigenetic mechanism. Histones accumulate their epigenetic information through post-translational chemical changes, such as acetylation, methylation, phosphorylation and ubiquitination, which take place in conserved amino acids. The modification of histones has a direct effect on gene transcription. In normal cells, the acetylation of certain lysines in the histones is related with genomic regions in active transcription (or euchromatin). On the contrary, methylation of specific residues of lysine or arginine in histones is associated with a transcriptional repressive pattern (heterochromatic). In tumor cells, the patterns of acetylation and methylation are found altered. These alterations may lead to silencing of TSG, such as p21.

Oncogenic Viruses

As previously mentioned, viral oncogenic genes can modify the cellular genetic program, transforming a normal cell into a tumor cell. We will mention two examples:

• **Papillomavirus**: **E6 and E7 viral proteins** from papillomavirus (described in chapter 1) inactivate p53 and Rb, respectively. This promotes proliferation and may cause cervical, penile, anal and head and neck cancer.

• **Hepatitis C virus** (HCV): It is estimated that there are about 200 million people infected with this virus, which remains, most of the times, in a latent state. Its oncogenic capability is associated to chronic hepatocyte infection. The development of a hepatocarcinoma is related with the virus' capacity of surviving in the liver and the auto-regenerative ability of hepatocytes. The specific causes that induce uncontrolled cellular proliferation are

still unknown, but it is thought that there is a loss in the cell cycle control when infected hepatocytes proliferate.

Angiogenesis and Metastasis

Angiogenesis and metastasis are of great importance for a primary tumor to invade other organs and to expand throughout the body. The growth of a tumor requires the formation of new vessels that supply oxygen and nutrients to cells (**fig. 4**). In fact, the probability of a neoplastic cell of dividing is inversely proportional to the distance to its closest vessel. If the tumor does not get vascularized, it becomes necrotic or remains in a quiescent state. Angiogenesis facilitates the entry of tumor cells into circulation, and thus, their dissemination into other organs. Tumor cells acquire an angiogenic phenotype that induces the formations of new capillaries, but these new vessels are aberrant, tortuous, with excessive ramifications and microhemorrhages. The main factors involved in this process are the vascular endothelium growth factor (**VEGF**) and the fibroblast growth factor (**FGF**). The formation of new blood vessels requires various steps that occur in a sequential manner: initiation of angionegic response, rupture of the basal membrane and degradation of the extracellular matrix, migration and proliferation of endothelial cells, cell-to-cell and cell-to-matrix interaction, and vascular remodeling.

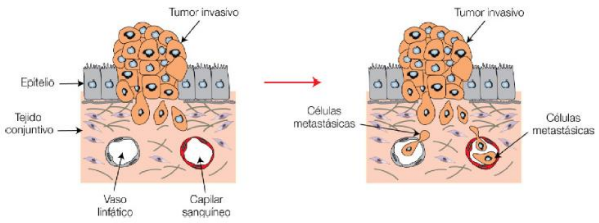


Fig. 4. Model of invasion and metastasis through blood vessels.

The main stimuli of the expression of VEGF are hypoxia and hypoglycemia, generally present in necrotic areas of the tumor. VEGF induces blood vessel permeability and stimulates the synthesis of proteins implicated in endothelial cell proliferation, migration and vascular remodeling.

Another essential step in tumor development is the invasion of peritumoral tissues by tumor cells: they acquire a migratory phenotype and destroy the extracellular matrix through the production of proteases. In invasive cells of epithelial tumors there is a decrease in **E-cadherin**, which facilitates the absence of adhesion with other epithelial cells and its mobility. These cells now acquire a mesenchymal phenotype, with expression of typical molecules of motile cells in the connective tissue. The capacity of producing metastasis is exclusive to malignant tumors and takes places, basically, through blood or lymphatic vessels. Angiogenesis plays an essential role in metastasization, since a tumor cell that is found near a vessel can intravasate and colonize another tissue. But not all cancerous cells found in blood are able to form a new tumor mass, because these cells are frequently destroyed by the immune system. In fact, the efficiency of colonization of a new organ is very low, and the majority of cells do not survive the adverse conditions that they face outside the primary tumor.

Dissemination can also occur through lymphatics. In order to know if metastasis exists in lymph nodes, the "sentinel node technique" is applied. This is the first lymph node that drains a territory. In order to detect which one it is, a radiotracer or a dye is injected into the tumor bed. This tracer will flow through lymphatics to the sentinel node. The histological study of this node allows to know whether this node is colonized by tumor cells. If a tumor has metastasized through lymphatics, it is most probable that it affects the sentinel node first on its path. If it is not affected, it is very likely that the rest of the nodes are not affected either.

Evasion from the Immune System

The immune system is continuously conducting an immunological vigilance to recognize altered or infected cells. In order to survive, tumor cells must evade the antitumoral response by the immune system. This explains that the incidence of cancer is higher in people with congenital immunodeficiency or immunosuppressed, such as patients with HIV or those treated with immunosuppressors (patients that suffered organ transplants).

Cancer cells produce tumor antigens that are not expressed by the rest of the cells. Thus, the protein products of oncogenes and mutated TSG can give rise to antigens that differ from those expressed by normal cells. These antigens can be processed and incorporated into the major histocompatibility complex class 1 (MHC class I) to be recognized by CD8+ T-lymphocytes. Phagocytosis of tumor antigens allows the presentation into MHC class II to be recognized by CD4+ T-lymphocytes.

But the organism does not always achieve tumor rejection. Sometimes, the immune response is too weak or the tumor proliferation rate exceeds the immune system's capacity to kill it. In addition, many tumors acquire systems to evade the immune system. These mechanisms are based on the decrease of the expression of MCH class I or presentation of tumor antigens. Neoplastic cells can also secrete immunosuppressor factors, such as TGF-β, which exert negative signals on cells of the immune system.

Cancer Treatments

Classic Treatments

The only available treatment before the 19th century was surgery. At the beginning of the 20th century, radiophysics and technological advances in ionizing radiation allowed the implementation of radiotherapy against cancer. This treatment uses photons or electrons in order to damage tumor cells, because it causes DNA breaks and generates large amount of free radicals that induce cell death in tumor cells.

By the mid-20th century, the first chemotherapy agents began to be developed, obtained from plants or derived from the nitrogenized mustard. The constant development of new drugs by the pharmaceutical industry has produced multitude of antineoplastic agents with the capacity to cure some types of cancers, to guarantee long periods of time until cancer relapse, to elongate life expectancy and to improve the quality of life (**fig. 5**).

Conventional chemotherapy agents kill mainly cells with high proliferation rates. These drugs show antitumoral activity, but they also kill normal cells that are proliferating. This lack of specificity is the main cause of side effects suffered by oncological patients. They damage hematopoietic precursors in bone marrow and produce pancytopenia (a decrease in red and white blood cells and platelets). They also damage intestine cells (in a constant renewal process), as well as hair follicles. In many cases, the toxicity exerted by conventional chemotherapy limits the use of these drugs in oncological patients.

Families of antineoplastic drugs

Anticancer drugs can be classified as follows: alkylanting agents, antimetabolites, mitotic inhibitors, topoisomerase inhibitors and antitumoral antibiotics.

ALKYLATING AGENTS: They were the first drugs employed in an efficient manner in cancer treatment. They are a heterogeneous group of drugs that react with DNA through an alkyl group, generating DNA breaks. The result is the induction of apoptosis or the halt of the cell cycle. Examples are **platinum derivatives** (i.e. cisplatin), which are included in most of the polychemotherapy regimes against cancer.

ANTIMETABOLITES: They are structural analogues to purines and pyrimidines that intervene naturally in the synthesis of nucleic acids. During DNA replication they are incorporated into the DNA and RNA, generating an altered sequence or giving rise to ruptures of the nucleic acids. An example is the **antipyrimidine 5-fluoroacyl, 5-FU**.

MITOTIC INHIBITORS: There are two groups of antineoplastic drugs within this family: vinca alkaloids (i.e. vincristine, vinblastine) and taxanes (i.e. taxol) used against many cancers. They interfere with microtubules during mitosis and cause cell death.

TOPOISOMERASE INHIBITORS: Topoisomerase inhibitors interfere with DNA replication.

ANTITUMORAL ANTIBIOTICS: This group of cytotoxic drugs was developed originally from Streptomyces and were conceived as antibiotic drugs. Later on, their cytotoxic activity against tumor cells was discovered. This group includes **anthracyclines**, frequently used in the treatment of breast cancer.

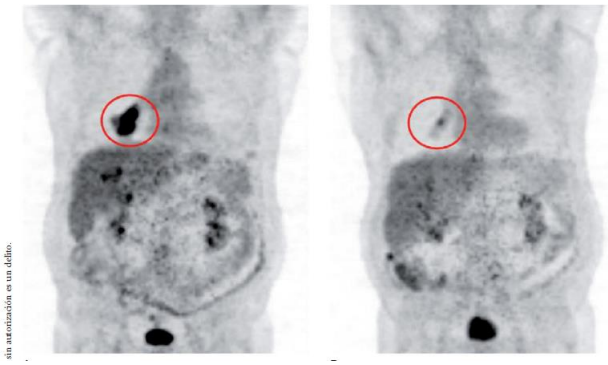


Fig. 5. PET imaging performed in a cancer patient. The black area highlighted in red corresponds to the hepatic area of the untreated tumor (on the left) or the tumor after treatment with chemotherapy (right).

Targeted therapy in cancer

In the last few years, our knowledge about molecular signaling pathways that regulate the growth of cancerous cells has increased considerably. Moreover, advances in biotechnology have permitted the synthesis of new sophisticated drugs, such as monoclonal antibodies and small molecules that can pass through the plasma membrane and block specific intracellular proteins. All this has contributed significantly to oncological treatments directed towards a specific "molecular target". This strategy looks for a better efficiency of treatments against cancer and a decrease in toxicity.

Classification of inhibitors

EGFR INHIBITORS: The inhibition of EGFR can take place by acting on its extracellular portion through monoclonal antibodies, or by blocking the tyrosine kinase activity of the receptor. **Gefitinib** is one of these drugs that inhibits the function of the receptor through blocking the ATP binding to the intracellular kinase domain.

HER-2 INHIBITORS: The strategies are also based on monoclonal antibodies or tyrosine kinase inhibitors. **Trastuzumab** and **pertuzumab** are two monoclonal antibodies that use slightly different mechanisms to treat breast cancer with amplification of the Her-2 gene. Nevertheless, the response rate to the monotherapy is limited and the clinical activity relies on combination with chemotherapeutics. Moreover, a percentage of patients does not respond or becomes resistant to the treatment.

VEGF AND VEGFR INHIBITORS: **Bevacizumab** is a pure antiangiogenic monoclonal antibody against VEGF that inhibits endothelial cell proliferation and tumor angiogenesis. It is indicated in combination with chemotherapy.